

ZECHARIAH DIEGELMAN

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Mechanical engineering senior at UNC Charlotte with hands-on experience in mechatronics, instrumentation, manufacturing, CAD, and technical analysis. Background in fabrication, construction, crew leadership, and customer-facing work; seeking internship or co-op opportunities in mechanical design, manufacturing, test, or project engineering.

EDUCATION

University of North Carolina at Charlotte - Charlotte, NC

Expected Fall 2027

Bachelor of Science in Mechanical Engineering | GPA: 3.11 | Senior

Relevant Coursework: Mechatronics I & II; Design Projects I; Manufacturing Systems; Thermodynamics I; Dynamic Systems I; Engineering Mechanics; Solid Mechanics; Engineering Graphics/CAD; Linear Algebra; Differential Equations

Current Coursework: Computational Methods; Fluid Mechanics; Mechanics & Materials Lab; Design Project II; Engineering Materials

Central Piedmont Community College - Charlotte, NC

July 2025

Associate in Engineering

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion 360, engineering drawings, shop drawings, dimensioning, standard/metric measurement

Programming & Analysis: MATLAB, LabVIEW, Microsoft Excel, technical data analysis, numerical problem solving, report writing

Electronics & Testing: Multisim, strain gauges, Wheatstone bridges, instrumentation amplifiers, sensors, data acquisition, circuit troubleshooting

Fabrication: Woodworking machinery, hand and power tools, prototyping, mechanical assembly, fit and finish inspection

SELECTED ENGINEERING PROJECTS

Vibrating Cantilever Beam Instrumentation & Dynamic Analysis

Summer 2026

Mechatronics II Projects Laboratory - UNC Charlotte

- Built and tested a strain-gauge Wheatstone bridge and AD623 instrumentation amplifier to measure beam deflection.
- Calibrated and acquired experimental data in LabVIEW; compared physical results with Multisim and theoretical models.
- Analyzed transient response, damping, resonance, and experimental error; documented findings in a technical memo.

Team Engineering Design Project

Fall 2025

Design Projects I Laboratory - UNC Charlotte

- Applied the engineering design process in a team setting, including requirements definition, concept evaluation, prototyping, testing, and final presentation.
- Used technical drawings, measurements, and iterative problem solving to refine the design and communicate results.

MATLAB Engineering Computation

Summer 2026

Computational Methods for Engineers - UNC Charlotte

- Developed scripts and functions to process engineering data, solve numerical problems, automate calculations, and verify outputs with test cases.

EXPERIENCE

Barker Brothers LLC - Landscaper & Marketing Assistant

May 2023-Present

Mint Hill, NC

- Construct and maintain patios, walkways, retaining walls, and landscape installations using measurements and hand/power tools.
- Support company marketing through website development, photography, video production, and social media content.

Mint Hill Cabinetry - Carpenter

Feb 2023-May 2023

Monroe, NC

- Interpreted cabinet designs and shop drawings to build and assemble high-end custom cabinetry.
- Operated table saws, dowel machines, edge banders, drills, saws, and nail guns while working from standard and metric dimensions.

LaborPanes - Crew Leader

Aug 2021-Jul 2022

Charlotte, NC

- Supervised crew members on residential job sites, quoted work, communicated with homeowners, and maintained vehicles and equipment.